

Sean Luka Girgis

Senior PySpark / Python Data Engineer

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Professional Summary

Senior Data Engineer with 20+ years of enterprise IT experience across Python, SQL, PySpark/Spark, Hadoop/Hive, ETL/ELT, distributed processing, data warehousing, performance optimization, and cloud data platform patterns. Strong background building scalable data pipelines, transformation frameworks, reporting datasets, telemetry analytics, forecasting pipelines, and governed data products using Python, SQL, PySpark, Pandas, Oracle, AWS S3/Glue/Athena/Redshift patterns, and production monitoring practices. Experienced collaborating with analytics, engineering, and business stakeholders to deliver reliable data solutions for enterprise reporting, advanced analytics, and machine-learning-supporting workflows.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built automated Python/Pandas ETL pipelines ingesting P95 telemetry from 6,000+ infrastructure endpoints, replacing manual processes with repeatable, validated data workflows.
- Used PySpark/Spark patterns to process large-scale telemetry and support parallel forecasting workloads across high-volume infrastructure datasets.
- Built hybrid AWS/Oracle data platform patterns using S3 landing zones, AWS Glue ETL concepts, Redshift analytics workloads, and Oracle schemas for historical retention and reporting.
- Designed and optimized SQL/Oracle reporting schemas to support data warehousing, capacity planning, trend analysis, executive dashboards, and operational risk reporting.
- Developed data integration, aggregation, and transformation logic to combine BMC TrueSight/TSCO, CA APM, and AppDynamics feeds into unified reporting datasets.
- Implemented data quality checks, reconciliation logic, and validation practices to improve confidence in enterprise performance and capacity reporting outputs.
- Developed Prophet and scikit-learn forecasting models to predict capacity bottlenecks months ahead and support ML-driven analytics initiatives.
- Created dashboards and reporting outputs using Streamlit, matplotlib, seaborn, and plotly to communicate performance, utilization, and capacity insights to stakeholders.
- Documented data definitions, transformations, validation rules, operational workflows, and platform decisions to support governance and knowledge transfer.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical hospitality systems, strengthening production visibility and performance analytics.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and provide actionable recommendations to engineering teams.
- Implemented End-User Monitoring and transaction tracing across web and mobile systems to improve visibility into user journeys, API behavior, and application performance.

- Integrated HP Performance Center with Dynatrace to correlate load-test behavior, APM telemetry, and infrastructure metrics.
- Participated in AWS cloud migration readiness discussions by evaluating monitoring coverage, operational risks, and data requirements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM subject matter expert for a large financial-services enterprise with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Designed dashboards, alerts, SLA thresholds, and management modules used to monitor enterprise application performance and support operational decision-making.
- Built Perl and Korn Shell data-extraction scripts to automate performance reporting and replace manual weekly exports.
- Partnered with J2EE/WebLogic and infrastructure teams to diagnose bottlenecks using APM telemetry, SQL/Oracle data, thread behavior, and infrastructure KPIs.
- Delivered technical documentation, onboarding guidance, and client training for monitoring standards, deployment processes, and production troubleshooting.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom web applications under load to identify throughput limits, response-time degradation, and resource bottlenecks including JDBC connections, threads, heap memory, CPU, and garbage collection.
- Used Java/J2EE, JMX monitoring, Wily Introscope, and thread-dump automation practices to provide real-time visibility during performance validation.
- Documented performance baselines, regression thresholds, findings, and recommendations used by engineering teams for release-gate decisions.
- Collaborated with application, testing, and infrastructure teams to troubleshoot performance defects and validate remediation outcomes.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while maintaining strict performance and reliability expectations.
- Optimized core transaction processing using C++, OCCI, OCI, SQL, and Oracle, reducing physical hardware footprint while preserving sub-second query latency.
- Built C++/OCCI/OCI validation tooling to verify migration correctness across large transaction datasets during phased cutover.
- Automated schema conversion and data validation workflows to protect data integrity across relational database migration paths.

Education

High Diploma / Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt / Cairo, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, C++, C, Java, Perl, Korn Shell, XML, JSON, Scala concepts

Flagship Projects

Serverless Lakehouse Platform (AWS)

Designed a serverless data platform pattern using S3, Glue, Athena, Redshift, Bedrock, PySpark, Parquet, and Snappy compression for cloud-scale ingestion, transformation, query, and AI-enabled access.

Technologies: AWS S3, AWS Glue, Athena, Amazon Redshift, PySpark, Parquet, Bedrock

HorizonScale — AI Capacity Forecasting Engine

Designed an AI-driven forecasting engine using Python, PySpark, Prophet, scikit-learn, and Streamlit to transform enterprise telemetry into capacity-risk predictions and operational decision support.

Technologies: Python, PySpark, Prophet, scikit-learn, Streamlit, Telemetry Data, Forecasting

Performance Telemetry and Capacity Reporting Platform

Built Python and SQL-driven reporting workflows to turn monitoring and infrastructure telemetry into dashboards, validated datasets, executive reports, and trend analysis.

Technologies: Python, SQL, Oracle, BMC TrueSight, AppDynamics, CA APM, Streamlit

AI-Powered Job Search Pipeline

Built an agentic Python pipeline for job intake, semantic duplicate detection, LLM scoring, JSON artifact generation, quality gates, and tracking using FAISS, sentence-transformers, Grok API, and Claude Code workflows.

Technologies: Python, FAISS, Sentence Transformers, LLM APIs, RAG, Agentic Workflows, JSON Contracts